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| **Fiscal Year:** | FY 2018 | **Task Last Updated:** | FY 08/30/2018 |
| **PI Name:** | Oates, Chris Ph.D. | | |
| **Project Title:** | Development/Evaluation of a 10^-17 Yb Optical Clock for Space Applications | | |
| **Division Name:** | Physical Sciences | | |
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| **Program/Discipline--Element/Subdiscipline:** | FUNDAMENTAL PHYSICS--Fundamental physics | | |
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| **Human Research Program Risks:** | None | | |
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| **Space Biology Cross-Element Discipline:** | None | | |
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| **Organization Name:** | National Institute of Standards and Technology | | |
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| **Zip Code:** | 80305-3337 | **Congressional District:** | 2 |
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| **No. of PhD Candidates:** | 3 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA JPL |
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| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Diddams, Scott Ph.D. ( National Institute of Standards & Technology )<br>Ludlow, Andrew Ph.D. ( National Institute of Standards & Technology ) | | |
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| **Task Description:** | The primary goal of the European Space Agency's (ESA) Space Optical Clock (SOC) program is to develop a high-performance optical frequency standard (clock) for operation in space with 1 x 10^-17 frequency uncertainty. National Institute of Standards and Technology (NIST) proposes to leverage its existing research on world-leading Yb lattice optical clocks to: (1) demonstrate a laboratory-based Yb lattice optical clock with frequency uncertainty of 1 x 10^-17 or better; (2) stabilize a space-friendly fiber frequency comb to the Yb clock signal; and (3) develop a preliminary design for space-qualification by providing clock subsystem specifications. The proposed program will provide ESA with detailed information to make the most informed decisions on use of one or more optical frequency standards for the flight portion of the SOC program. Currently, ESA has identified the Sr and Yb lattice optical clocks as the most promising technologies. A Yb lattice clock has significant potential advantages over Sr in that Yb clocks can operate with solid state lasers much more likely to be practical in space operations. Several European groups are presently evaluating Sr's space prospects and Sr has thus received significant attention. For Yb, however, the most mature and high performance lattice clock resides in our lab at NIST. This Yb clock currently performs at least as well as the world's best Sr clocks, approximately 10^-16 frequency uncertainty in all cases, and continues to rapidly improve. Thus, we are uniquely qualified to play a vital role in the SOC project. For a modest investment, the proposed research program will ensure ESA has the best information and options to select the optical clock technology to meet SOC program goals, ensuring fully informed decisions affecting a much larger financial and scientific investment in the SOC program. Furthermore, by demonstrating a 10^-17 level Yb optical lattice clock, our proposed effort will achieve significant milestones currently outlined in the SOC program. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The NIST SOC program (Yb lattice clock) also directly supports the major benefits of the overall Atomic Clock Ensemble in Space (ACES) program by developing a new high performance NIST optical frequency standard that will directly contribute to NIST's roles in the broader ACES program:<br>1. Substantial improvements in international coordination of extremely precise and accurate time and frequency information. Precision timekeeping and synchronization are crucial for such ubiquitous technologies as precision positioning and navigation, telecommunications, remote sensing, surveillance, and many other applications. The current technical limitations on international time and frequency transfer impose fundamental limits on these technologies. This project seeks to improve international time and frequency transfer by at least a factor of 10, with the potential for significantly greater improvements. Such improvements would not only directly enhance the current technologies depending on precision timing and synchronization, but would also open up new applications directly related to ultraprecise timing such as gravimetry, geodesy, magnetometry, and other measurements based on tiny perturbations in precision frequency measurements. The NIST Yb lattice clock represents a dramatic improvement in stability and accuracy compared to cesium microwave primary frequency standards.<br><br>2. This project will serve as a pathfinder demonstration of the ability to perform cold atom experiments in space. Cold atoms and molecules represent a major, growing research opportunity for a wide variety of technologies, from ultraprecision measurements of all kinds (time, gravity, magnetic fields, electrical quantities, temperature, etc.) to the potential for development of exotic new materials. Having mastered cold atom and molecule experiments in the Earth laboratory, being able to perform such experiments in space opens the benefits of precision measurements in the space environment as well as microgravity experiments not possible in the Earth-based laboratory.<br><br>3. Fundamental physics experiments. The operation of a cold atom clock in space provides the opportunity to significantly improve measurements on the limits of special and general relativity and related phenomena, search for possible Lorentz transformation variations, and search for possible time variations in fundamental constants, at levels 10 to 100 times more precise than previous experiments. Most major physics discoveries in recent decades have come from tiny variations in expected behavior, and these experiments would have profound implications on fundamental physics and cosomology. |
| | Progress in the most recent reporting period (April 2017 through May 2018) has been substantial. The two key aspects of improving an atomic clock's performance are to make the ticking more regular (stability) and to make the ticking rate match more closely the reference atom's natural ticking rate (termed uncertainty or accuracy). The main goal of this proposal was to demonstrate a fractional uncertainty of less than one part in 10^17 for the Yb lattice clock. Achieving such an ambitious goal required that we evaluate all the possible effects that could shift the clock frequency away from that of the atom and then verify this level of performance through clock comparisons. We note that our continued efforts to improve the stability as well has greatly accelerated this investigation. During the previous year, we completed an exhaustive Yb lattice clock evaluation with a result that is a factor of five improvement beyond the stated goal. Over the past year we have performed the most precise/accurate clock comparison for atomic clocks to date, which achieved a precision of 2.3 x 10^-19 and agreement between two virtually independent Yb clocks below 1 part in 10^18, a factor of ten beyond our original goal. The major advances for this reporting period include:<br>1. Verification of 10^-18 performance for the Yb clock<br><br>(a) Completion of the uncertainty budget for the Yb lattice clock<br><br>In the past year we completed an exhaustive evaluation of the Yb lattice clock system, which came in with a total clock fractional frequency uncertainty of 1.5 x 10^-18, the lowest value reported to date for an atomic clock. The uncertainty was limited principally by the lattice light shift, a by-product of the trapping light used to confine the Yb atoms in the optical lattice, which is formed by a tightly focused standing wave of light. A detailed study of the light shift was required to reach the achieved level and was reported on last year in Physical Review Letters. A second key element to achieving this low uncertainty was the use of a room temperature copper Faraday shield to reduce the shift due to the influence of blackbody radiation on the clock energy levels. The relevant technology and theory in our experimental apparatus was described in a second article in Physical Review Letters, which reported a fractional uncertainty of 3 x 10^-19 for the blackbody shift when the atomic sample is confined inside the copper enclosure. The enclosure also shields the system from stray dc electric fields at the sub-10^-19 level. (b) Completion of a second Yb lattice clock<br><br>To verify predicted uncertainty budgets for new clock systems, it is essential to perform comparisons between independently constructed versions at a precision beyond the predicted levels. To this end, in the past year we completed construction of a similar, though not identical Yb lattice clock. This system uses a similar blackbody shield, which is necessary to achieve 10^-18 performance in a room temperature system. For this second system, named Yb-2, |

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| **Task Progress:** | we use a build-up cavity for the lattice light, which both helps manage the light shift better, but also gives a different enough system to guard against repeating a possible mistake with this critical shift that might otherwise be common mode with that of the original system, Yb-1. An evaluation of this second system yielded a slightly smaller fractional frequency uncertainty of 1.4 x 10^-18.<br><br>(c) High precision comparisons between the two Yb lattice clocks<br><br>Benefiting from the excellent control over systematic shifts that can compromise instability on long timescales, we were able to reach new levels for measurement instability for atomic clock comparisons. For one long run taken over 72 hours (with 88% uptime), the comparison revealed a single-clock instability of 3.2 x 10^-19, the first clock instability below 10^-18. For a series of ten runs taken over the course of more than a month, the frequency difference between the two systems was measured to be 7(9) x 10^-19, the highest degree of reproducibility to date for atomic clocks. We note that the uncertainty of this measurement was slightly less than that of the standards themselves because certain theoretical systematics (e. g., uncertainties in values of atomic constants) are common to the two systems. We emphasize that this level of uncertainty is well below the uncertainty of the correction for the gravitational redshift of clocks in Boulder, CO (presently this number is 6 x 10^-18). In fact, an uncertainty of 1 x 10^-18 corresponds to a 1 cm change in height of an atomic sample used in the clock apparatus. While future refinements to existing geoid models will reduce the uncertainties, levels at or below 1 cm appear unlikely in the near future, especially considering mean sea level rises of 3.3(3) mm/year. As a result, optical clocks may one day become precise tools for beyond state-of-the-art geodesic measurements.<br><br>(d) Measuring the absolute frequency of the Yb clock transition<br><br>As the atomic clock community prepares for a future redefinition of the SI Second with an optical transition (we note the above Yb comparison shows agreement with an uncertainty 100 times below that of the best Cs microwave reproducibility), it has laid out a series of five milestones that must be accomplished in advance of such a redefinition. One key step is to have absolute frequency measurements at the limit of the performance of today's microwave primary standards to ensure continuity between current and future definitions. To this end, we have run a comparison between the Yb standards and the NIST Timescale over the past 8 months. Because the NIST Timescale is continually linked to Bureau of Weights and Measures (BIPM), which is in turn linked to Primary Standards around the world, we have been able to derive an absolute value for the Yb optical clock transition with an uncertainty of ~2 x 10^-16, limited by the primary standards themselves. The resultant value is in excellent agreement with a newly recommended value by BIPM, which was based on a combination of previous Yb measurements and Sr/Yb ratio measurements. This agreement basically shows a consistency between the frequencies of the Yb optical clocks, the Sr optical clocks, and the Cs microwave clocks from around the world, with the "loop" closed at the limit of the microwave sources' performance. This represents the most stringent realization to date of key milestones on the road to redefinition.<br><br>2. Testing fundamental physics with Yb lattice clocks<br><br>Finally, the NIST Yb clock is presently taking part in two different efforts to search for dark matter, by placing constraints on dark/light matter coupling strengths. The first is an international collaboration between two laboratories in Europe, one in Japan, and the Yb clock at NIST. Each lab monitors frequency excursions between the atom-stabilized and optical-cavity-stabilized laser frequency of each optical clock. Correlations between distinct laboratory clocks operating simultaneously can be used to constrain coupling strengths. The second effort focuses on the optical frequency ratio measurements taking place here at NIST, where the distinct sensitivity of each atomic species to the fine structure constant is exploited to place constraints on possible dark matter coupling. Together with our collaborators, optical clock data from both efforts is being analyzed. Furthermore, we continue to make additional optical frequency ratio measurements at NIST to add data for these dark matter search efforts. Early analysis indicates that significant improvements in constraints on the coupling to ultralight scalar dark matter, as well as topological defect dark matter, are possible with the existing clock measurements. |

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|---|---|
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